
This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Video Projector
Manufacturer: Christie
Firmware Version: N/A
Model(s): DHD951-Q, DWU951, DWX951, DXG1051

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
3.01.0000-b010	2.1.0

Version History

Module Version	Date	Notes
1_0_4_0	12/19/2018	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`
- If login credentials are required, devicePassword must be set accordingly.
Example: `InterfaceName.devicePassword = 'extron'`

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='DHD951-Q')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='DHD951-Q')</code>
EthernetClass
<code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 23, Model='DHD951-Q')</code>

Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AspectRatio	Value 'Normal' '16:10'	Value '4:3' '14:9'	Value '16:9' 'Native'
# AspectRatio example InterfaceName.Set('AspectRatio', 'Normal')			
Command AutoImage	Value None		
# AutoImage example InterfaceName.Set('AutoImage', None)			
Command Blank	Value 'On'	Value 'Off'	
# Blank example InterfaceName.Set('Blank', 'On')			
Command BlankColor	Value 'My Screen' 'White'	Value 'Original' 'Black'	Value 'Blue'
# BlankColor example InterfaceName.Set('BlankColor', 'My Screen')			
Command ClosedCaption	Value 'Off'	Value 'On'	
# ClosedCaption example InterfaceName.Set('ClosedCaption', 'Off')			
Command ClosedCaptionChannel	Value '1' – '4'		
# ClosedCaptionChannel example InterfaceName.Set('ClosedCaptionChannel', '1')			
Command ClosedCaptionMode	Value 'Captions'	Value 'Text'	
# ClosedCaptionMode example InterfaceName.Set('ClosedCaptionMode', 'Captions')			
Command EcoMode	Value 'Normal'	Value 'Eco'	
# EcoMode example InterfaceName.Set('EcoMode', 'Normal')			

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Command ExecutiveMode	Value 'On'	Value 'Off'	
# ExecutiveMode example InterfaceName.Set('ExecutiveMode', 'On')			
Command Freeze	Value 'Off'	Value 'On'	
# Freeze example InterfaceName.Set('Freeze', 'Off')			
Command Input	Value 'PC' 'HDMI 2' 'VIDEO'	Value 'LAN' 'DVI-D' 'PC 2'	Value 'HDMI 1' 'HDBaseT' 'SDI/DIGITAL 1'
# Input example InterfaceName.Set('Input', 'PC')			
Command LampMode	Value 'Dual' 'Alternate'	Value 'Lamp 1'	Value 'Lamp 2'
# LampMode example InterfaceName.Set('LampMode', 'Dual')			
Command LensMemoryIndex	Value '1' – '3'		
# LensMemoryIndex example InterfaceName.Set('LensMemoryIndex', '1')			
Command LensMemoryLoad	Value None		
# LensMemoryLoad example InterfaceName.Set('LensMemoryLoad', None)			
Command OnScreenDisplay	Value 'Off'	Value 'On'	Value 'Hide'
# OnScreenDisplay example InterfaceName.Set('OnScreenDisplay', 'Off')			
Command PictureMode	Value 'Standard' 'Dynamic' 'WhiteBoard' 'User 1'	Value 'Natural' 'BlackBoard' 'Daytime' 'User 2'	Value 'Cinema' 'GreenBoard' 'Dicom' 'User 3'
# PictureMode example InterfaceName.Set('PictureMode', 'Standard')			
Command Power	Value 'Off'	Value 'On'	
# Power example			

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InterfaceName.Set('Power', 'Off')		
Command	Value	Value
Shutter	'On'	'Off'
# Shutter example InterfaceName.Set('Shutter', 'On')		

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command AspectRatio	Value 'Normal'	Value '4:3'	Value '16:9'
	'16:10'	'14:9'	'Native'
# AspectRatio examples InterfaceName.Update('AspectRatio') Value = InterfaceName.ReadStatus('AspectRatio') InterfaceName.SubscribeStatus('AspectRatio', None, FeedbackHandler)			
Command Blank	Value 'On'	Value 'Off'	
# Blank examples InterfaceName.Update('Blank') Value = InterfaceName.ReadStatus('Blank') InterfaceName.SubscribeStatus('Blank', None, FeedbackHandler)			
Command BlankColor	Value 'My Screen'	Value 'Original'	Value 'Blue'
	'White'	'Black'	
# BlankColor examples InterfaceName.Update('BlankColor') Value = InterfaceName.ReadStatus('BlankColor') InterfaceName.SubscribeStatus('BlankColor', None, FeedbackHandler)			
Command ClosedCaption	Value 'Off'	Value 'On'	
# ClosedCaption examples InterfaceName.Update('ClosedCaption') Value = InterfaceName.ReadStatus('ClosedCaption') InterfaceName.SubscribeStatus('ClosedCaption', None, FeedbackHandler)			
Command	Value		

ClosedCaptionChannel	'1' – '4'		
# ClosedCaptionChannel examples InterfaceName.Update('ClosedCaptionChannel') Value = InterfaceName.ReadStatus('ClosedCaptionChannel') InterfaceName.SubscribeStatus('ClosedCaptionChannel', None, FeedbackHandler)			
Command	Value	Value	
ClosedCaptionMode	'Captions'	'Text'	
# ClosedCaptionMode examples InterfaceName.Update('ClosedCaptionMode') Value = InterfaceName.ReadStatus('ClosedCaptionMode') InterfaceName.SubscribeStatus('ClosedCaptionMode', None, FeedbackHandler)			
Command	Value	Value	
ConnectionStatus	'Connected'	'Disconnected'	
# ConnectionStatus examples Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command	Value	Value	Value
DeviceStatus	'Cover Error'	'Fan Error'	'Lamp Error'
	'Temp Error'	'Air flow Error'	'Cold Error'
	'Filter Error'	'Shutter Error'	'Lens Shift Error'
	'Lamp 1 Warning'	'Lamp 2 Warning'	'Humidity Error'
	'Color Wheel Error'	'Dynamic Iris Error'	'Other Error'
	'Normal'		
# DeviceStatus examples InterfaceName.Update('DeviceStatus') Value = InterfaceName.ReadStatus('DeviceStatus') InterfaceName.SubscribeStatus('DeviceStatus', None, FeedbackHandler)			
Command	Value	Value	
EcoMode	'Normal'	'Eco'	
# EcoMode examples InterfaceName.Update('EcoMode') Value = InterfaceName.ReadStatus('EcoMode') InterfaceName.SubscribeStatus('EcoMode', None, FeedbackHandler)			
Command	Value	Value	
ExecutiveMode	'On'	'Off'	
# ExecutiveMode examples InterfaceName.Update('ExecutiveMode') Value = InterfaceName.ReadStatus('ExecutiveMode') InterfaceName.SubscribeStatus('ExecutiveMode', None, FeedbackHandler)			
Command	Value		
FilterUsage	0 – 2000		
# FilterUsage examples InterfaceName.Update('FilterUsage') Value = InterfaceName.ReadStatus('FilterUsage')			

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InterfaceName.SubscribeStatus('FilterUsage', None, FeedbackHandler)			
Command Freeze	Value 'Off'	Value 'On'	
# Freeze examples InterfaceName.Update('Freeze') Value = InterfaceName.ReadStatus('Freeze') InterfaceName.SubscribeStatus('Freeze', None, FeedbackHandler)			
Command Input	Value 'PC' 'HDMI 2' 'VIDEO'	Value 'LAN' 'DVI-D' 'PC 2'	Value 'HDMI 1' 'HDBaseT' 'SDI/DIGITAL 1'
# Input examples InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)			
Command LampMode	Value 'Dual' 'Alternate'	Value 'Lamp 1'	Value 'Lamp 2'
# LampMode examples InterfaceName.Update('LampMode') Value = InterfaceName.ReadStatus('LampMode') InterfaceName.SubscribeStatus('LampMode', None, FeedbackHandler)			
Command LampUsage	Value Hours		
Qualifier Key 'Lamp'	Qualifier Value '1' – '2'		
# LampUsage examples InterfaceName.Update('LampUsage', {'Lamp': '1'}) Value = InterfaceName.ReadStatus('LampUsage', {'Lamp': '1'}) InterfaceName.SubscribeStatus('LampUsage', None, FeedbackHandler)			
Command LensMemoryIndex	Value '1' – '3'		
# LensMemoryIndex examples InterfaceName.Update('LensMemoryIndex') Value = InterfaceName.ReadStatus('LensMemoryIndex') InterfaceName.SubscribeStatus('LensMemoryIndex', None, FeedbackHandler)			
Command OnScreenDisplay	Value 'Off'	Value 'On'	Value 'Hide'
# OnScreenDisplay examples InterfaceName.Update('OnScreenDisplay') Value = InterfaceName.ReadStatus('OnScreenDisplay') InterfaceName.SubscribeStatus('OnScreenDisplay', None, FeedbackHandler)			
Command PictureMode	Value 'Standard'	Value 'Natural'	Value 'Cinema'

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	'Dynamic'	'BlackBoard'	'GreenBoard'
	'WhiteBoard'	'Daytime'	'Dicom'
	'User 1'	'User 2'	'User 3'
<pre># PictureMode examples InterfaceName.Update('PictureMode') Value = InterfaceName.ReadStatus('PictureMode') InterfaceName.SubscribeStatus('PictureMode', None, FeedbackHandler)</pre>			
Command Power	Value 'Off'	Value 'On'	Value 'Cooling Down'
<pre># Power examples InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)</pre>			
Command Shutter	Value 'On'	Value 'Off'	
<pre># Shutter examples InterfaceName.Update('Shutter') Value = InterfaceName.ReadStatus('Shutter') InterfaceName.SubscribeStatus('Shutter', None, FeedbackHandler)</pre>			

Cable and Adapter Requirements

Captive Screw to Female DB9 RS-232 Serial Cable

Notes for the Device

Serial communication

Port Type: RS-232

Baud Rate: 19200

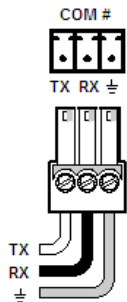
Data Bits: 8

Parity: None

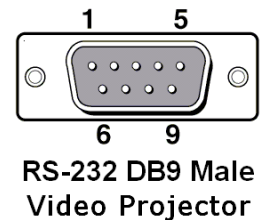
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD		3	TxD
RxD		2	RxD
GND		5	GND



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scripter ethernet interface

Port Type:	Ethernet
Default Port:	23
Logon Credentials Supported:	Yes
Multi-Connection Capabilities:	Undetermined
Port Changeability:	No

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Appendix A. Set Commands

Aspect Ratio 14:9	\xBE\xEF\x03\x06\x00\xCE\xD6\x01\x00\x08 \x09\x00
Aspect Ratio 16:10	\xBE\xEF\x03\x06\x00>\xD6\x01\x00\x08 \x0A\x00
Aspect Ratio 16:9	\xBE\xEF\x03\x06\x00\x0E\xD1\x01\x00\x08 \x01\x00
Aspect Ratio 4:3	\xBE\xEF\x03\x06\x00\x9E\xD0\x01\x00\x08 \x00\x00
Aspect Ratio Native	\xBE\xEF\x03\x06\x00^\xD7\x01\x00\x08 \x08\x00
Aspect Ratio Normal	\xBE\xEF\x03\x06\x00^\xDD\x01\x00\x08 \x10\x00
Auto Image None	\xBE\xEF\x03\x06\x00\x91\xD0\x06\x00\x0A \x00\x00
Blank Color Black	\xBE\xEF\x03\x06\x00\x9B\xD0\x01\x00\x000\x06\x00
Blank Color Blue	\xBE\xEF\x03\x06\x00\xCB\xD3\x01\x00\x000\x03\x00
Blank Color My Screen	\xBE\xEF\x03\x06\x00\xFB\xCA\x01\x00\x000 \x00
Blank Color Original	\xBE\xEF\x03\x06\x00\xFB\xE2\x01\x00\x000@\x00
Blank Color White	\xBE\xEF\x03\x06\x00k\xD0\x01\x00\x000\x05\x00
Blank Off	\xBE\xEF\x03\x06\x00\xFB\xD8\x01\x00 0\x00\x00
Blank On	\xBE\xEF\x03\x06\x00k\xD9\x01\x00 0\x01\x00
Closed Caption Channel 1	\xBE\xEF\x03\x06\x00\xD2b\x01\x00\x027\x01\x00
Closed Caption Channel 4	\xBE\xEF\x03\x06\x00\x82a\x01\x00\x027\x04\x00
Closed Caption Mode Captions	\xBE\xEF\x03\x06\x00\x06c\x01\x00\x017\x00\x00
Closed Caption Mode Text	\xBE\xEF\x03\x06\x00\x96b\x01\x00\x017\x01\x00
Closed Caption Off	\xBE\xEF\x03\x06\x00\xFAB\x01\x00\x007\x00\x00
Closed Caption On	\xBE\xEF\x03\x06\x00jc\x01\x00\x007\x01\x00
Eco Mode Eco	\xBE\xEF\x03\x06\x00\xAB"\x01\x00\x003\x01\x00
Eco Mode Normal	\xBE\xEF\x03\x06\x00;#\x01\x00\x003\x00\x00
Executive Mode Off	\xBE\xEF\x03\x06\x00;\xEF\x01\x00\xC00\x00\x00
Executive Mode On	\xBE\xEF\x03\x06\x00\xAB\xEE\x01\x00\xC00\x01\x00
Freeze Off	\xBE\xEF\x03\x06\x00\x83\xD2\x01\x00\x020\x00\x00
Freeze On	\xBE\xEF\x03\x06\x00\x13\xD3\x01\x00\x020\x01\x00
Input DVI-D	\xBE\xEF\x03\x06\x00\xAE\xD4\x01\x00\x00 \x09\x00
Input HDBaseT	\xBE\xEF\x03\x06\x00\xAE\xDE\x01\x00\x00 \x11\x00
Input HDMI 1	\xBE\xEF\x03\x06\x00\x0E\xD2\x01\x00\x00 \x03\x00
Input HDMI 2	\xBE\xEF\x03\x06\x00n\xD6\x01\x00\x00 \x0D\x00
Input LAN	\xBE\xEF\x03\x06\x00\xCE\xD5\x01\x00\x00 \x0B\x00
Input PC	\xBE\xEF\x03\x06\x00\xFE\xD2\x01\x00\x00 \x00\x00
Input PC 2	\xBE\xEF\x03\x06\x00>\xD0\x01\x00\x00 \x04\x00
Input SDI/DIGITAL 1	\xBE\xEF\x03\x06\x00^\xDE\x01\x00\x00 \x12\x00
Input VIDEO	\xBE\xEF\x03\x06\x00n\xD3\x01\x00\x00 \x01\x00
Lamp Mode Alternate	\xBE\xEF\x03\x06\x00\xDF, \x01\x00\x0B3\x10\x00

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Lamp Mode Dual	\xBE\xEF\x03\x06\x00\x1F!\x01\x00\x0B3\x00\x00
Lamp Mode Lamp 1	\xBE\xEF\x03\x06\x00\x8F \x01\x00\x0B3\x01\x00
Lamp Mode Lamp 2	\xBE\xEF\x03\x06\x00\x7F \x01\x00\x0B3\x02\x00
Lens Memory Index 1	\xBE\xEF\x03\x06\x00K\x92\x01\x00\x07\$\x00\x00
Lens Memory Index 3	\xBE\xEF\x03\x06\x00+\x93\x01\x00\x07\$\x02\x00
Lens Memory Load None	\xBE\xEF\x03\x06\x00\xE8\x90\x06\x00\x08\$\x00\x00
On Screen Display Hide	\xBE\xEF\x03\x06\x00\xEF\xD7\x01\x00\x170\x02\x00
On Screen Display Off	\xBE\xEF\x03\x06\x00\x8F\xD6\x01\x00\x170\x00\x00
On Screen Display On	\xBE\xEF\x03\x06\x00\x1F\xD7\x01\x00\x170\x01\x00
Picture Mode BlackBoard	\xBE\xEF\x03\x06\x00\xE3\xEF\x01\x00\xBA0 \x00
Picture Mode Cinema	\xBE\xEF\x03\x06\x00\xB3\xF7\x01\x00\xBA0\x01\x00
Picture Mode Daytime	\xBE\xEF\x03\x06\x00\xE3\xC7\x01\x00\xBA0@\x00
Picture Mode Dicom	\xBE\xEF\x03\x06\x00s\xC6\x01\x00\xBA0A\x00
Picture Mode Dynamic	\xBE\xEF\x03\x06\x00\xE3\xF4\x01\x00\xBA0\x04\x00
Picture Mode GreenBoard	\xBE\xEF\x03\x06\x00s\xEE\x01\x00\xBA0!\x00
Picture Mode Natural	\xBE\xEF\x03\x06\x00#\xF6\x01\x00\xBA0\x00\x00
Picture Mode Standard	\xBE\xEF\x03\x06\x00\x83\xF5\x01\x00\xBA0\x06\x00
Picture Mode User 1	\xBE\xEF\x03\x06\x00\xE3\xFB\x01\x00\xBA0\x10\x00
Picture Mode User 2	\xBE\xEF\x03\x06\x00s\xFA\x01\x00\xBA0\x11\x00
Picture Mode User 3	\xBE\xEF\x03\x06\x00\x83\xFA\x01\x00\xBA0\x12\x00
Picture Mode WhiteBoard	\xBE\xEF\x03\x06\x00\x83\xEE\x01\x00\xBA0"\x00
Power Off	\xBE\xEF\x03\x06\x00*\xD3\x01\x00\x00` \x00\x00
Power On	\xBE\xEF\x03\x06\x00\xBA\xD2\x01\x00\x00` \x01\x00
Shutter Off	\xBE\xEF\x03\x06\x00\xF3\x93\x01\x00\x05\$\x00\x00
Shutter On	\xBE\xEF\x03\x06\x00c\x92\x01\x00\x05\$\x01\x00

Appendix B. Update Commands

Aspect Ratio	\xBE\xEF\x03\x06\x00\xAD\xD0\x02\x00\x08 \x00\x00
Blank	\xBE\xEF\x03\x06\x00\xC8\xD8\x02\x00 0\x00\x00
Blank Color	\xBE\xEF\x03\x06\x00\x08\xD3\x02\x00\x000\x00\x00
Closed Caption	\xBE\xEF\x03\x06\x00\xC9b\x02\x00\x007\x00\x00
Closed Caption Channel	\xBE\xEF\x03\x06\x00qc\x02\x00\x027\x00\x00
Closed Caption Mode	\xBE\xEF\x03\x06\x005c\x02\x00\x017\x00\x00
Device Status	\xBE\xEF\x03\x06\x00\xD9\xD8\x02\x00 ` \x00\x00
Eco Mode	\xBE\xEF\x03\x06\x00\x08#\x02\x00\x003\x00\x00
Executive Mode	\xBE\xEF\x03\x06\x00\x08\xEF\x02\x00\xC00\x00\x00
Filter Usage	\xBE\xEF\x03\x06\x00\xC2\xF0\x02\x00\xA0\x10\x00\x00
Freeze	\xBE\xEF\x03\x06\x00\xB0\xD2\x02\x00\x020\x00\x00
Input	\xBE\xEF\x03\x06\x00\xCD\xD2\x02\x00\x00 \x00\x00
Lamp Mode	\xBE\xEF\x03\x06\x00,!\x02\x00\x0B3\x00\x00
Lamp Usage Lamp 1	\xBE\xEF\x03\x06\x00\xC2\xFF\x02\x00\x90\x10\x00\x00
Lamp Usage Lamp 2	\xBE\xEF\x03\x06\x00\x02\xAE\x02\x00\x90\x11\x00\x00
Lens Memory Index	\xBE\xEF\x03\x06\x00x\x92\x02\x00\x07\$\x00\x00
On Screen Display	\xBE\xEF\x03\x06\x00\xBC\xD6\x02\x00\x170\x00\x00
Picture Mode	\xBE\xEF\x03\x06\x00\x10\xF6\x02\x00\xBA0\x00\x00
Power	\xBE\xEF\x03\x06\x00\x19\xD3\x02\x00\x00` \x00\x00
Shutter	\xBE\xEF\x03\x06\x00\xC0\x93\x02\x00\x05\$\x00\x00